

HARRISBURG, PA, USA

WMO: 723990

Lat: 40.196N

Lon: 76.772W

Elev: 95

StdP: 100.19

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14711

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.0	-9.6	-19.5	0.7	-10.1	-17.3	0.8	-8.1	12.9	-1.4	12.1	-1.9	4.4	300	0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	33.2	24.0	32.0	23.3	30.4	22.7	25.5	30.9	24.7	29.6	24.0	28.5	3.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.1	28.3	23.0	18.0	27.2	22.5	17.4	26.6	78.7	30.9	75.3	29.6	72.1	28.6	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.5	9.9	8.5	DB	-15.0	35.7	4.0	1.7	-17.8	36.9	-20.1	37.9	-22.4	38.9	-25.2	40.2	(4)
(5)			WB	-15.7	27.1	3.7	1.0	-18.4	27.9	-20.6	28.5	-22.7	29.0	-25.4	29.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec				
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)				
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.4	-0.9	0.6	5.4	11.9	17.4	22.5	25.0	23.9	20.1	13.4	7.1	2.1	(6)		
(7)		DBStd	9.95	5.38	4.97	5.02	4.51	4.17	3.38	2.72	2.66	3.62	4.29	4.35	4.49	(7)		
(8)		HDD10.0	1162	339	264	158	29	1	0	0	0	14	107	248		(8)		
(9)		HDD18.3	2792	596	496	400	198	69	7	0	1	23	164	337	502	(9)		
(10)		CDD10.0	2056	2	2	17	87	231	375	464	432	302	119	21	4	(10)		
(11)		CDD18.3	644	0	0	1	6	41	132	207	174	75	10	0	0	(11)		
(12)		CDH23.3	5277	0	1	10	75	353	1067	1863	1352	494	62	1	0	(12)		
(13)		CDH26.7	1789	0	0	2	16	105	353	719	443	140	11	0	0	(13)		
(14)	Wind	WSAvg	3.2	3.8	3.9	4.2	3.9	3.2	2.9	2.6	2.4	2.5	2.8	3.3	3.5	(14)		
(15)	Precipitation	PrecAvg	1033	72	74	83	81	111	96	92	85	89	73	88	82	(15)		
(16)		PrecMax	1506	204	151	155	202	247	471	247	197	380	251	183	192	(16)		
(17)		PrecMin	764	11	14	26	11	13	2	21	24	16	1	20	20	(17)		
(18)		PrecStd	174	44	36	32	41	56	83	52	41	73	45	41	47	(18)		
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.0	18.0	24.1	28.8	32.1	33.8	36.3	34.2	32.6	28.0	21.4	18.1	(19)		
(20)			MCWB	14.2	12.1	15.5	18.4	22.0	24.1	24.3	24.4	23.1	21.0	15.3	14.6	(20)		
(21)		2%	DB	12.5	13.3	19.6	25.7	29.7	32.1	33.9	32.3	30.1	24.9	18.8	14.0	(21)		
(22)			MCWB	9.6	8.3	12.9	17.0	21.3	23.6	24.2	23.9	22.2	19.3	14.5	11.2	(22)		
(23)		5%	DB	8.6	10.1	16.3	22.5	27.3	30.3	32.2	30.7	28.0	22.5	16.3	10.9	(23)		
(24)			MCWB	6.0	6.2	10.9	15.1	19.8	22.6	23.6	23.0	21.2	18.1	12.3	8.6	(24)		
(25)		10%	DB	6.3	7.4	13.3	20.0	24.9	28.7	30.6	29.0	26.2	20.4	14.0	8.4	(25)		
(26)			MCWB	4.0	4.4	8.7	13.6	18.4	21.8	23.0	22.3	20.2	16.5	10.7	6.0	(26)		
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.0	14.2	16.5	20.1	23.3	25.8	26.7	26.7	25.0	22.8	17.7	15.2	(27)		
(28)			MADB	16.4	16.9	23.8	25.3	29.5	31.3	32.8	32.0	29.9	25.7	19.7	17.2	(28)		
(29)		2%	WB	10.2	9.0	14.0	18.1	22.0	24.7	25.6	25.4	23.7	20.9	15.5	12.1	(29)		
(30)			MADB	11.9	12.6	18.2	23.2	27.9	30.0	31.3	29.9	27.7	23.8	17.8	13.7	(30)		
(31)		5%	WB	6.7	6.6	11.7	16.2	20.8	23.8	24.8	24.5	22.7	19.0	13.5	8.7	(31)		
(32)			MADB	8.2	9.3	15.5	21.3	25.8	28.6	30.1	28.6	26.3	21.6	15.6	10.3	(32)		
(33)		10%	WB	4.2	4.8	9.2	14.5	19.6	22.9	24.0	23.6	21.6	17.1	11.3	6.4	(33)		
(34)			MADB	6.0	7.4	13.1	19.0	23.7	27.2	28.8	27.5	24.9	19.7	13.5	8.2	(34)		
(35)	Mean Daily Temperature Range			MDBR	7.7	8.6	9.6	10.8	10.3	9.8	9.7	9.5	9.8	9.9	8.9	7.4	(35)	
(36)		5% DB			MADB	10.8	13.1	14.3	14.9	13.3	11.6	11.4	10.9	11.8	12.5	12.4	10.9	(36)
(37)					MCWBR	8.7	9.2	8.9	7.9	6.5	5.3	4.5	4.7	5.6	7.2	8.3	8.8	(37)
(38)		5% WB			MADB	10.3	11.5	13.2	13.0	11.7	10.4	9.9	9.6	9.7	10.3	10.3	9.7	(38)
(39)					MCWBR	9.0	8.9	9.1	8.0	6.3	5.4	4.6	4.8	5.3	6.6	8.0	8.7	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.312	0.330	0.370	0.400	0.431	0.502	0.513	0.501	0.420	0.371	0.330	0.310	(40)		
(41)		taud		2.424	2.348	2.317	2.267	2.214	2.051	2.036	2.054	2.252	2.395	2.475	2.470	(41)		
(42)		Ebn at Noon		849	881	880	874	851	788	775	771	816	825	823	823	(42)		
(43)		Edh at Noon		81	101	116	131	141	166	168	160	123	96	77	73	(43)		
(44)	All-Sky Solar Radiation	RadAvg		1.82	2.71	3.63	4.71	5.28	5.85	5.90	5.16	4.22	3.00	2.11	1.55	(44)		
(45)		RadStd		0.15	0.34	0.27	0.44	0.62	0.49	0.32	0.28	0.55	0.34	0.20	0.18	(45)		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+85	N/A
(47) Regional (63 neighbors)		+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+119	+65

Nomenclature: See separate page